

CSE4203: Computer Graphics
Lecture – 2

Raster Graphics

**Fundamentals of Computer
Graphics, 4th Edition Chapter 3**

Outline

- Raster and Raster Images
- Image Compression
- Display Devices
- Pixel Values
- RGB Color
- Alpha Compositing

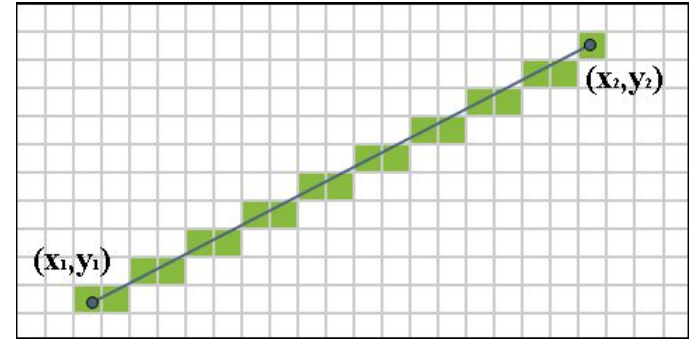
Raster (1/1)

- Most computer graphics images are presented on *raster display*.

- i.e. television

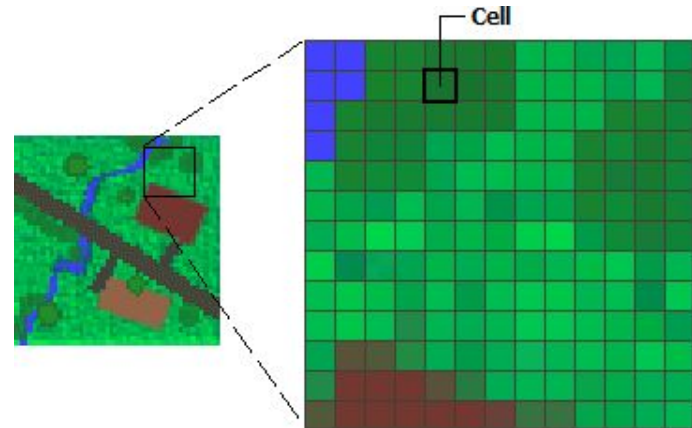
- has rectangular array of small light-emitting **pixels**

- individually set to different colors to create desired image.



Storing Images (1/2)

- Raster Image:
 - used to store and process images, as rasters are common in devices
- simply a **2D array**
- stores the pixel value for each pixel
- usually a color stored as **three numbers (r, g, b)**



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>
Image source: <https://desktop.arcgis.com/en/arcmap/10.3/manage-data/raster-and-images/what-is-raster-data.htm>

Storing Images (2/2)

- Raster Image:
 - Considered as device-independent of the image to be displayed
 - Resolution Dependent i.e quality are measured using number of pixels per unit such as DPI
 - Resizing can result quality degradation
 - Common formats are JPEG, PNG, BMP

Image Compression (1/4)

- Image compression is a method used to **reduce the size of images**,
- Improves the rendering speed with reduced file size
- 2 methods of compression:
 - Lossless Compression
 - Lossy Compression

Image Compression (2/4)

Lossless Compression

- Don't impact the image quality
- Only removes additional, non-essential data automatically added by the device used to take the photo
- No significant reduction in file size
- Lossless Compression algorithms: Run-length encoding, Huffman coding
- Lossless formats are .RAW, .BMP, .GIF, and .PNG

Image Compression (3/4)

Lossy Compression

- Reduces the file size considerably by removing image data
- Quality might degrade
- This process is irreversible - can't get back to the original file
- Common algorithms - discrete wavelet transform, fractal compression, transform encryption etc.
- Lossless format; JPEG, MPEG, AVC

Image Compression (4/4)

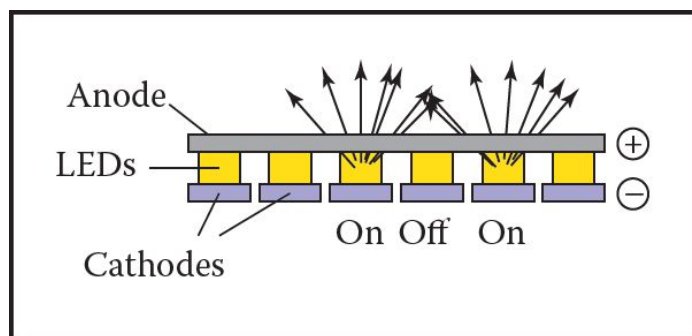


Display Devices (1/1)

- Transmissive Displays:
 - require a light source to illuminate them
 - backlight behind the array
 - i.e. in a projector, a lamp emits light projected onto the screen after passing through the array.
- Emissive Display:
 - it is its own light source.

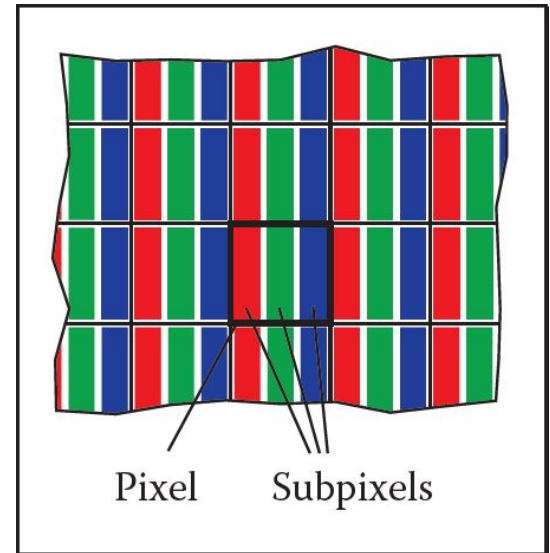
Emissive Displays (1/2)

- Emissive Displays:
 - Example: light-emitting diode (LED)
 - Each pixel is composed of one or more LEDs (semiconductor devices)
 - emit light with intensity $\leftrightarrow$ electrical current passing through them



Emissive Displays (2/2)

- Sub-pixel:
 - Pixels divided into three independently controlled sub-pixels (R, G, B)
 - each with own LED (different materials)
 - emit light of different colors

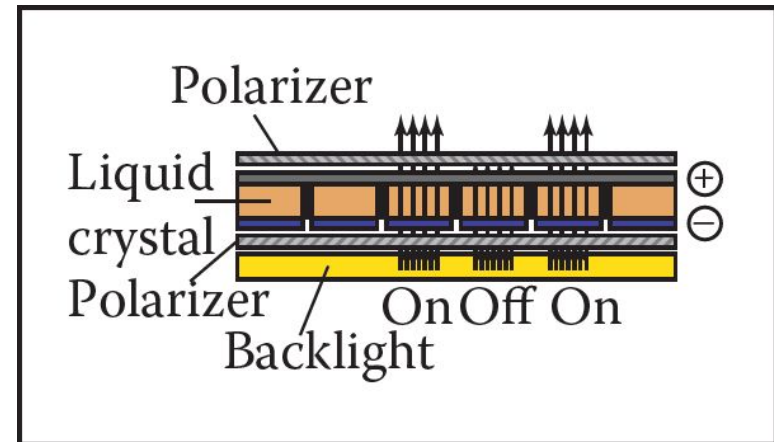


Transmissive Displays (1/3)

- Transmissive Displays:

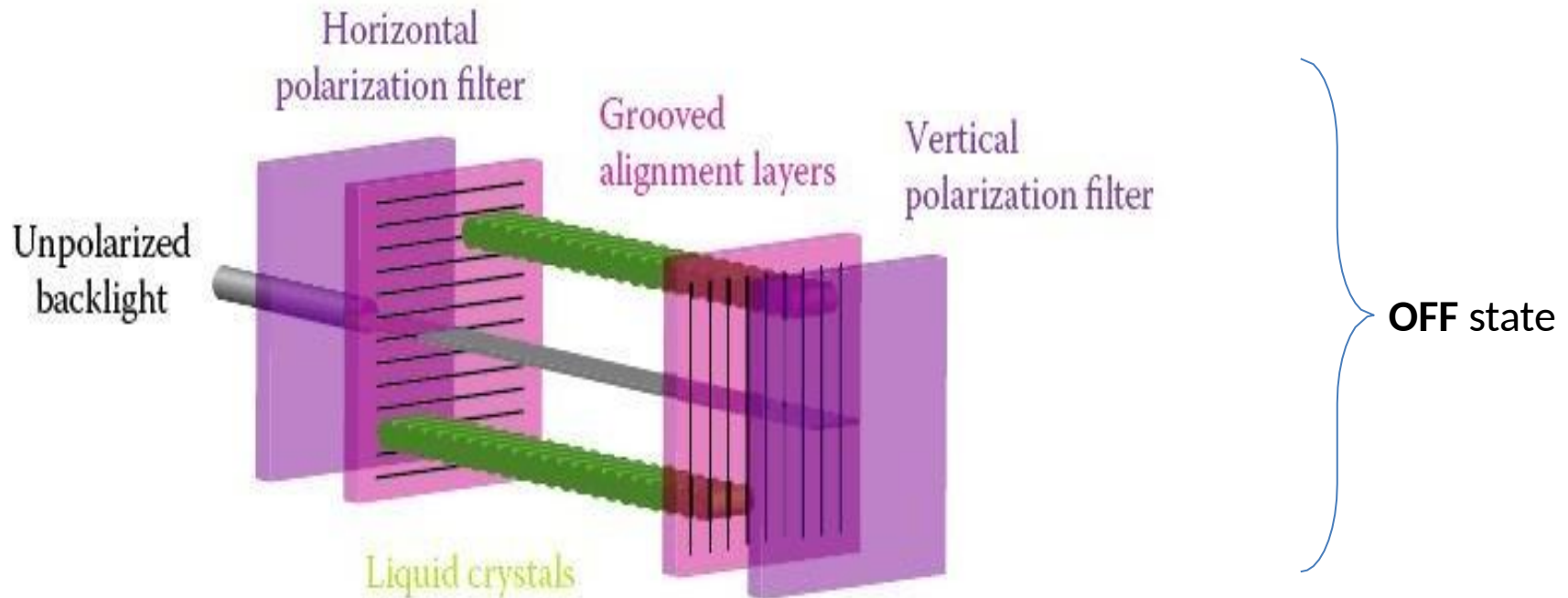
- Example: light crystal display (LCD)

- Molecular structure of liquid crystal rotates the polarization of light that passes through it
 - LCDs also have sub-pixels.



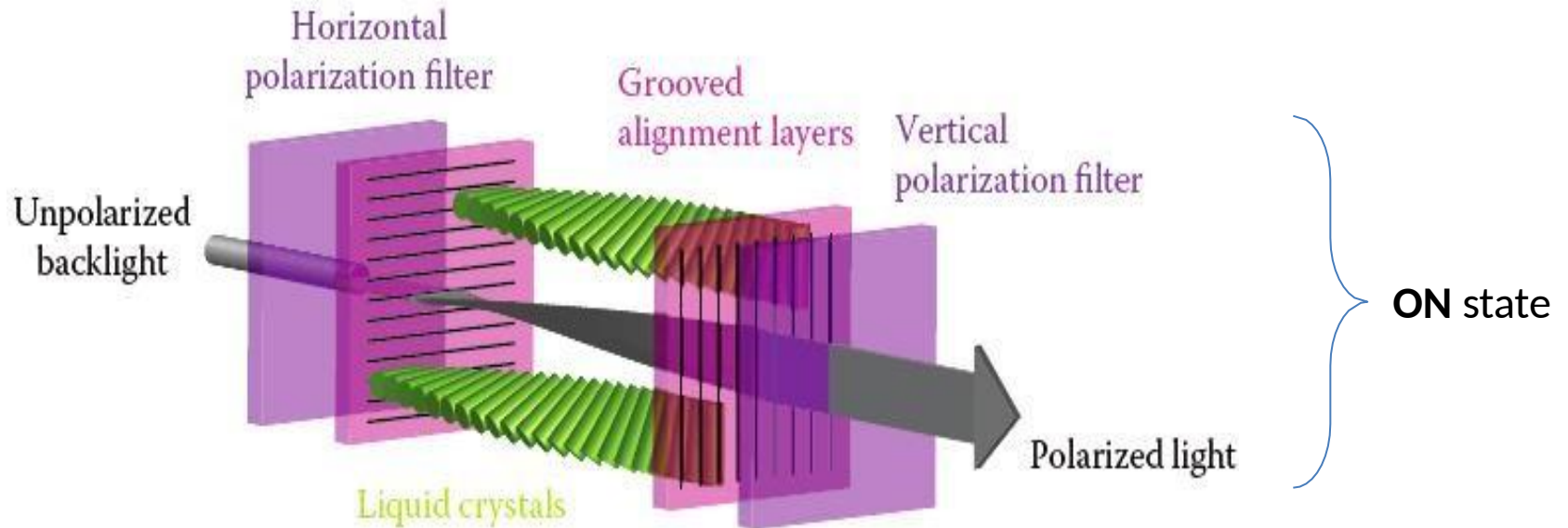
Transmissive Displays (2/3)

- Degree of rotation $\leftrightarrow$ applied voltage



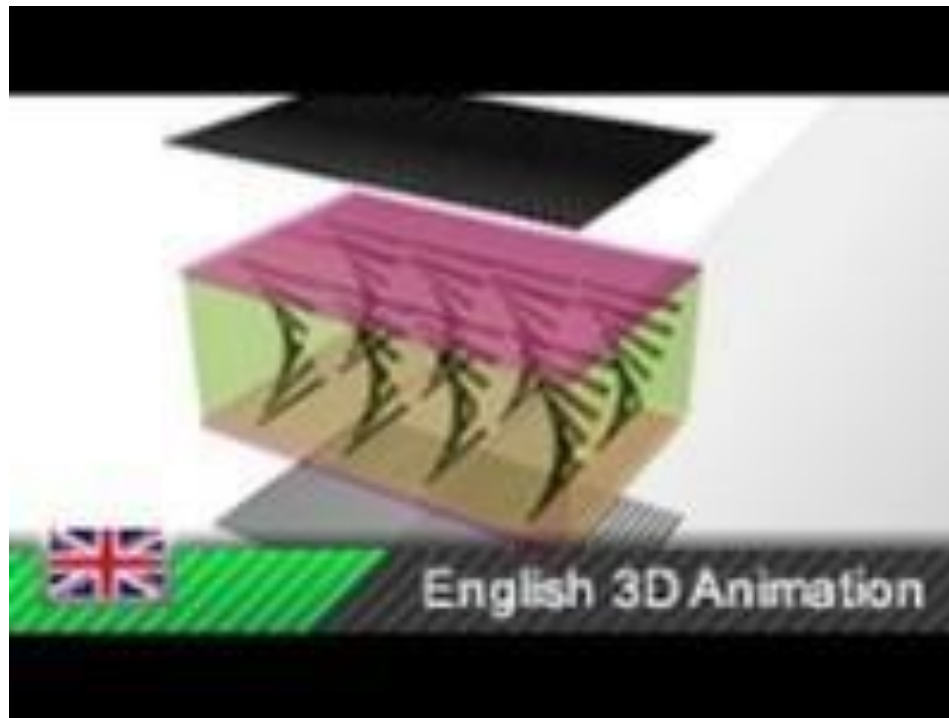
Transmissive Displays (2/3)

- Degree of rotation $\leftrightarrow$ applied voltage



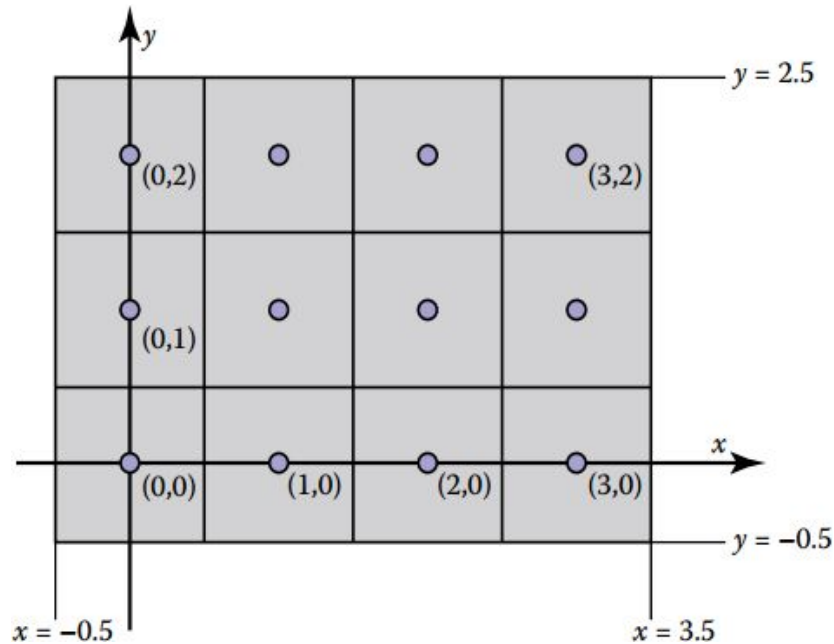
Transmissive Displays (3/3)

youtu.be/k7xGQKpQAWw?t=77



Pixel Values (1/1)

- Coordinate system for raster screen:
 - Convention:



RGB Color (1/1)

black = (0, 0, 0), red = (1, 0, 0), green = (0, 1, 0),
blue = (0, 0, 1), yellow = (1, 1, 0), magenta = (1, 0, 1),
cyan = (0, 1, 1), white = (1, 1, 1)

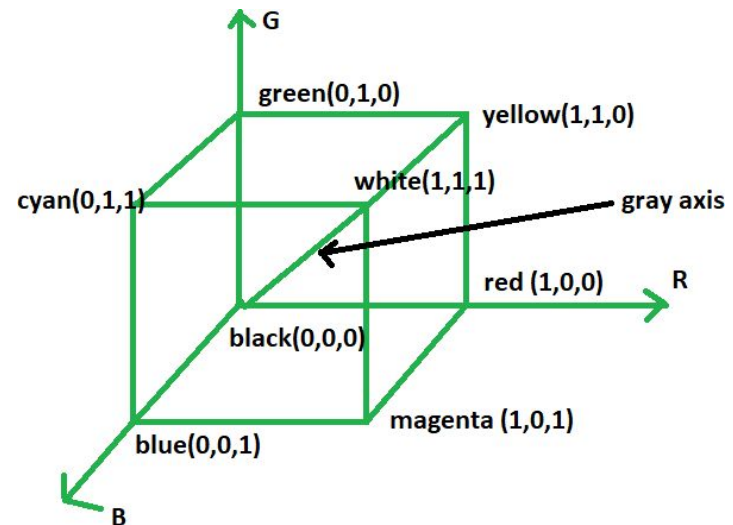
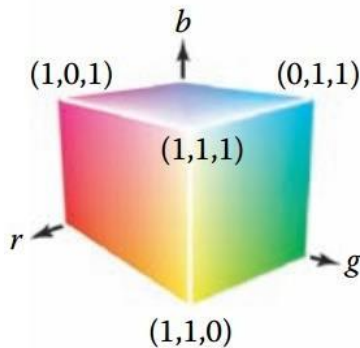
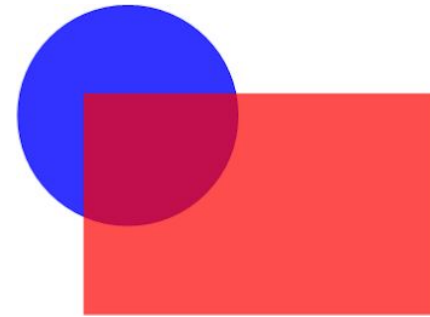


Image Source: <https://www.geeksforgeeks.org/computer-graphics-the-rgb-color-model/>

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

Alpha Compositing (1/3)

- Partially overwriting the contents of a pixel.
 - Where we have a background and want to insert a foreground image over it.
 - Transparent
 - Opaque (not transparent)
 - Partially Transparent



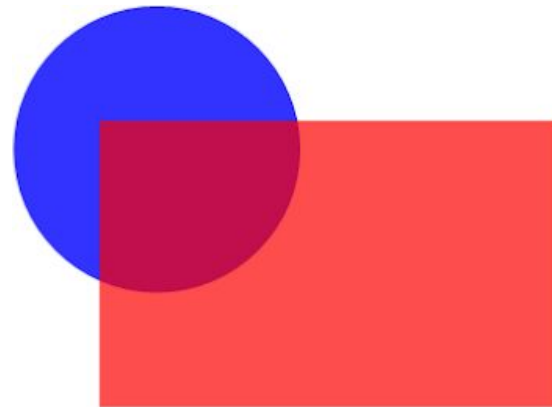
Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>
Image source: <http://www.graphicalweb.org/2005/papers/abstractsvgopen/index.html>

Alpha Compositing (2/3)

- foreground and background must be blended.

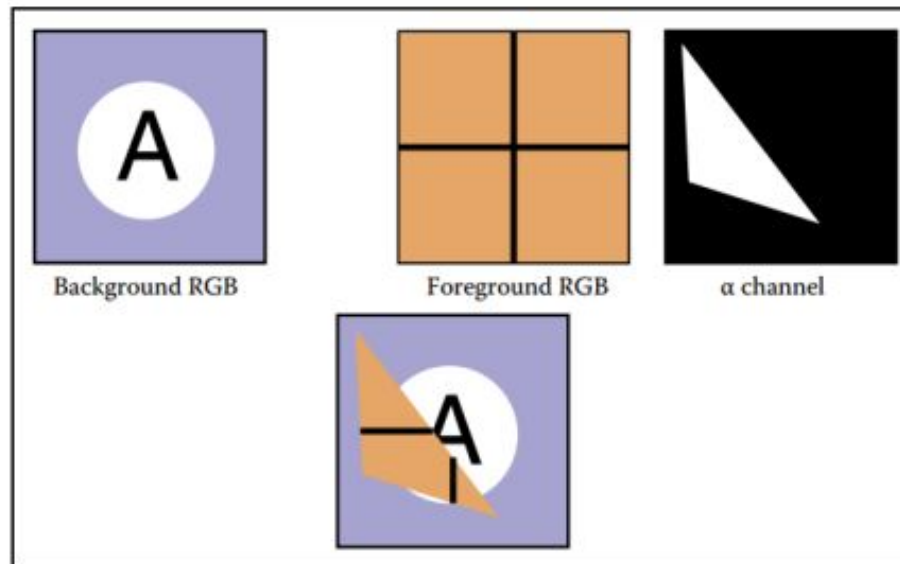
$$c = \alpha c_f + (1 - \alpha)c_b$$

- α = Fraction of the pixel covered by the foreground layer

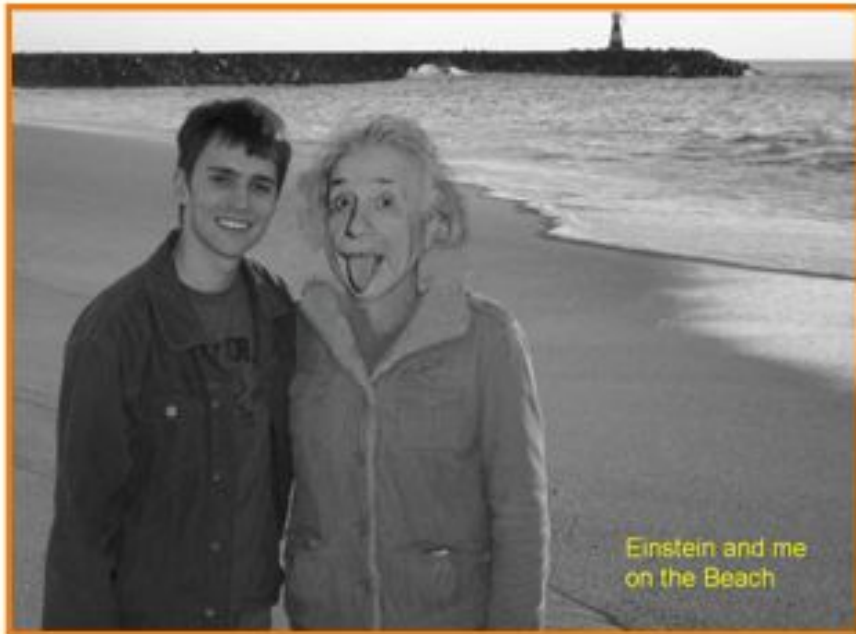


Alpha Compositing (3/3)

- Alpha Mask:
 - The α values for all the pixels is stored in a separate gray scale image.



What can you do with it? Get a photo with Einstein



What can you do with it? Horror Photo!!



Practice Question 1

- Given that, $C_f = 1.0$, $C_b = 0.5$ and $C = 0.8$, where, C_f , C_b and C are the foreground, background and composite intensities respectively. Determine the alpha(α) value to perform alpha compositing.

Further Reading

- Fundamentals of Computer Graphics, 4th Edition -
Chapter 3
- <https://www.adobe.com/uk/creativecloud/photography/discover/lossy-vs-lossless.html>

Thank You